

Satyam Chatrola

satyamchatrola14@gmail.com

[Portfolio](#)

[LinkedIn](#)

[GitHub](#)

[DeepDives](#)

PROFESSIONAL SUMMARY

AI Systems Engineer specializing in **low-latency inference & real-time serving**: profiling and cutting p99 tail latency, engineering async concurrency & backpressure for throughput under load, and hardening distributed pipelines. Deepening into **LLM serving internals**: KV-cache efficiency, roofline analysis, and correctness under optimization.

SKILLS

Inference & Serving: NVIDIA Triton, vLLM, SGLang, LlamaCpp, TensorRT, ONNX, OpenVINO

Systems & Real-Time: Python (async I/O, native C/Rust bindings), Asyncio, gRPC, WebRTC, LiveKit

Infrastructure & MLOps: Kubernetes (EKS), Docker, CI/CD (GitHub Actions, ArgoCD), CouchDB, PostgreSQL

Observability & Data: Datadog, Langfuse, Honeycomb, Prometheus, Grafana

AI & Modeling: PyTorch, Transformers, LLMs, Hugging Face, TensorFlow, MLFlow, FAISS, Pinecone

EXPERIENCE

Mursion, AI Engineer (Systems, Real-Time) March 2026 - Present

- **Cut p99 latency from 395 ms to 13 ms** (mean loop lag -71%) in production by offloading blocking Krisp C-extension calls off the asyncio event loop via `asyncio.to_thread`, preserving session ordering
- **Sustained throughput under heavy concurrent load** by replacing blocking loggers with async logging queue with high/low-watermark load-shedding and eventual-consistency backpressure, minimizing GIL contention
- **Accelerated telemetry serialization 5x** with orjson and made requests traceable via canonical IDs tagged as Datadog facets (CPU, memory, Garbage Collection pauses), propagated to Langfuse for AI-trace correlation
- **Root-caused and resolved a production traffic-skew incident** (recognized by engineering leadership) by isolating silent random misrouting between two builds during a Jenkins to ArgoCD migration
- **Shrank deploy footprint from 14 GB to 355 MB** & PR-to-prod from **30 to 7 min** with multi-stage Docker builds

Rapidops, Machine Learning Engineer May 2020 - June 2023 & October 2025 - March 2026

- Engineered Agentic AI workflows via self-hosted models (**SGLang/vLLM**) with tracing and observability
- Built a Two-Tower + BM25 retrieval system at 5M+ SKUs, with DeepFM ranking at **0.87 NDCG@10** and **7.2% lift**
- Optimized features and model quality across production pipelines, driving a **34% CTR lift** and **\$300K+ ARR**

PROJECTS

Care Companion: High-Performance MLOps Pipeline for Real-Time Inference [\[GitHub\]](#)

- Engineered a Triton inference service (P90 250ms) with **TensorRT (GPU)** and **OpenVINO (CPU)** **ONNX** backends
- Built a Prometheus/Grafana observability stack tracking live **data drift** (KL divergence) and **model degradation**

TECHNICAL FEATURES AND ARTICLES

- **Quoted by name in** [Qualcomm's developer blog](#) (On-Device AI Builders Hackathon - Qualcomm x LM Studio x Microsoft, hosted at NYU Tandon) on the power efficiency of on-device LLM inference.
- "A Pragmatic Engineer's Guide to OOMs, NCCL, and Resilience in Large-Scale Training" [\[View Article\]](#)
- "The Transformer Architecture: A First-Principles Analysis from Self-Attention to KV Caching" [\[View Article\]](#)

EDUCATION

New York University - MS in Computer Science (CGPA: 3.70)

September 2023 - May 2025

Gujarat Technological University - BE in Computer Engineering (CGPA: 3.79)

June 2018 - June 2022